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P40 MAE 5730
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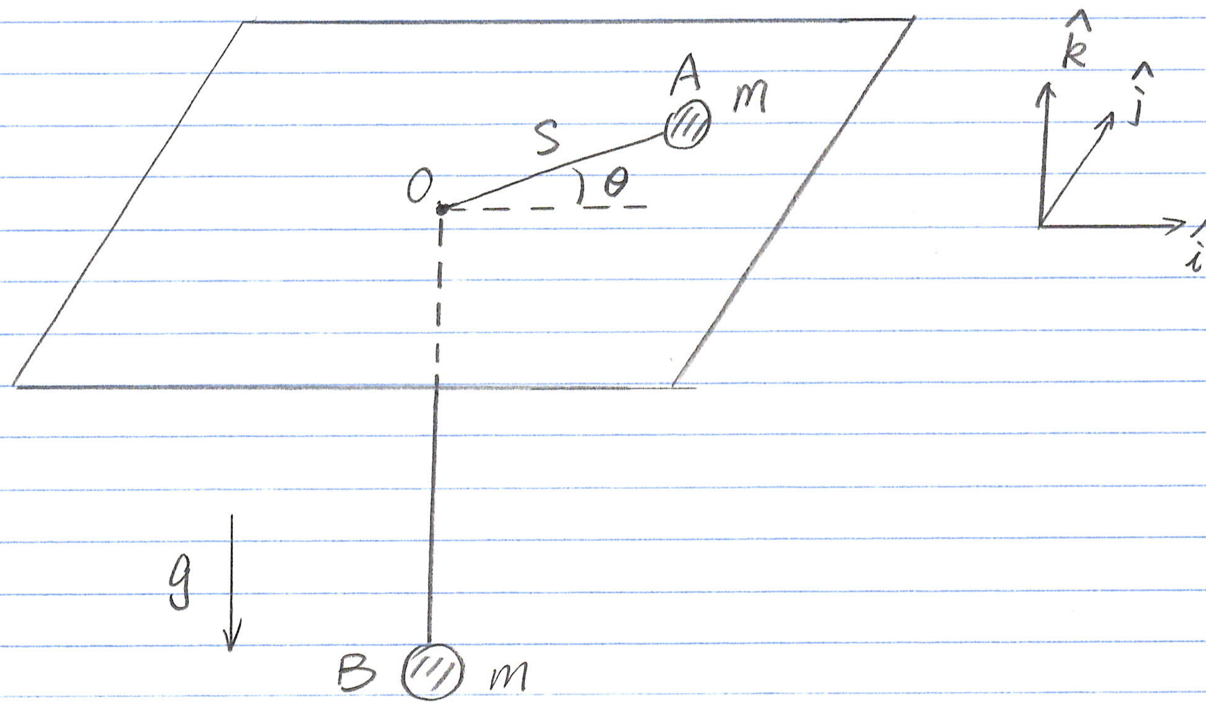
Q-exam Dynamics

Problem 1

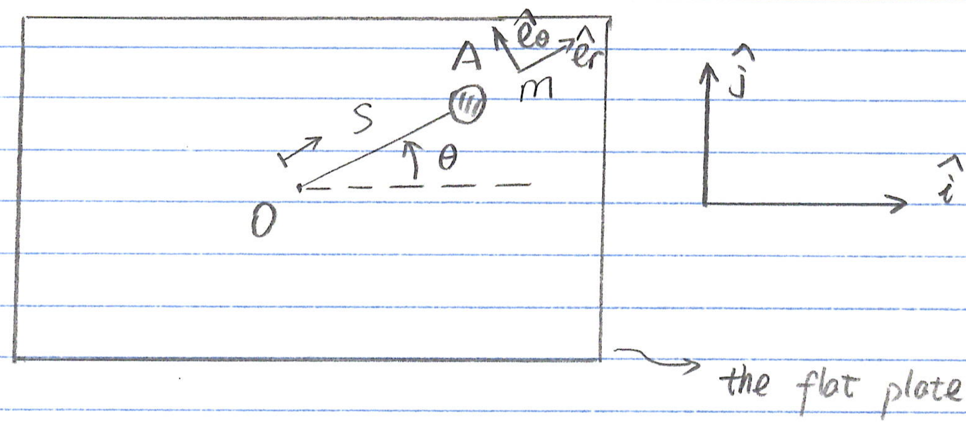


(a) How many degrees of freedom?
(b) Are there any conserved quantities?
(c) Find the equations of motion.

Assumptions:
① The plate is flat;
② No friction;
③ The string's length = constant.
④ Two masses equal.

Solution:

(a) This system has TWO degrees of freedom : θ and S



(b) Total Energy $E_{tot} = E_K + E_P$ is conserved ;
The $\vec{H}_{/O}$ is conserved.

(c) $\vec{r}_{A/O} = s\,\hat{e}_r$

$$\Rightarrow \vec{v}_{A/O} = \frac{d}{dt}(\vec{r}_{A/O}) = \dot{s}\,\hat{e}_r + s\dot{\theta}\,\hat{e}_\theta$$

Since the string's length is assumed to be constant, then we have

$$\vec{v}_{B/O} = \dot{s}\,\hat{k}$$

Thus, we can compute E_K & E_P

$$E_K = \frac{1}{2} m v_{A/O}^2 + \frac{1}{2} m v_{B/O}^2$$

$$= \frac{1}{2} m (\dot{s}^2 + s^2\dot{\theta}^2) + \frac{1}{2} m \dot{s}^2$$

$$= m\dot{s}^2 + \frac{1}{2} m s^2 \dot{\theta}^2$$

$$E_P = mgs$$

$$\mathcal{L} = E_k - E_p = m\dot{s}^2 + \frac{1}{2}ms^2\dot{\theta}^2 - mgs$$

s:

$$\frac{\partial \mathcal{L}}{\partial \dot{s}} = 2m\dot{s} \quad , \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{s}} = 2m\ddot{s}$$

$$\frac{\partial \mathcal{L}}{\partial s} = ms\dot{\theta}^2 - mg$$

Thus, $\quad 2m\ddot{s} - ms\dot{\theta}^2 + mg = 0$

or,

$$\ddot{s} - \frac{1}{2}s\dot{\theta}^2 + \frac{1}{2}g = 0$$

θ:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = ms^2\dot{\theta} \quad , \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) = ms^2\ddot{\theta} + 2ms\dot{s}\dot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0$$

Thus, $\quad ms^2\ddot{\theta} + 2ms\dot{s}\dot{\theta} = 0$

or,

$$\ddot{\theta} + 2\frac{\dot{s}}{s}\dot{\theta} = 0$$

To conclude,

$$\begin{cases} \ddot{s} = \frac{1}{2}s\dot{\theta}^2 - \frac{1}{2}g \\ \ddot{\theta} = -2\frac{\dot{s}}{s}\dot{\theta} \end{cases}$$

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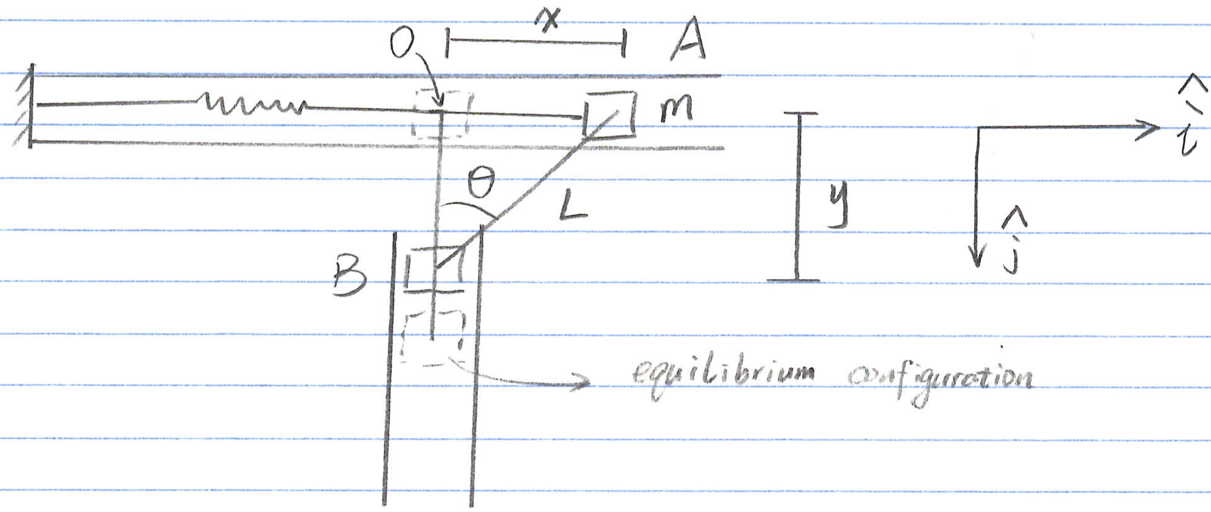


Derive the equation of motion

Assumptions:

L = rigid rod

neglect friction

include gravity.

Solution:

1 DoF: θ

$$x = L\sin\theta, \quad y = L\cos\theta$$
$$\dot{x} = L\dot{\theta}\cos\theta, \quad \dot{y} = -L\dot{\theta}\sin\theta$$

$$E_k = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2 = \frac{1}{2}m\left[L^2\dot{\theta}^2\cos^2\theta + L^2\dot{\theta}^2\sin^2\theta\right]$$
$$= \frac{1}{2}mL^2\dot{\theta}^2$$

$$E_p = -mgy + \frac{1}{2}kx^2 = -mgL\cos\theta + \frac{1}{2}kL^2\sin^2\theta$$

$$\mathcal{L} = E_k - E_p = \frac{1}{2}mL^2\dot{\theta}^2 + mgL\cos\theta - \frac{1}{2}kL^2\sin^2\theta$$

$$\frac{\partial\mathcal{L}}{\partial\dot{\theta}} = mL^2\dot{\theta}, \quad \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right) = mL^2\ddot{\theta}$$
$$\frac{\partial\mathcal{L}}{\partial\theta} = -mgL\sin\theta - kL^2\sin\theta\cos\theta$$

Thus

$$mL^2\ddot{\theta} + mgL\sin\theta + kL^2\sin\theta\cos\theta = 0$$

or,

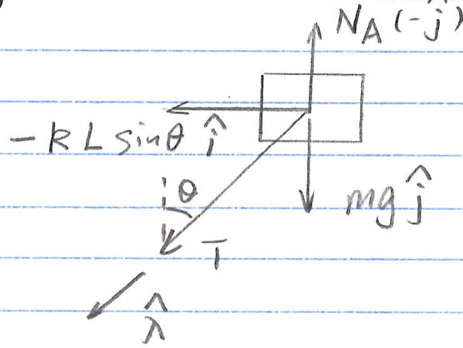
$$\ddot{\theta} + \frac{g}{L}\sin\theta + \frac{k}{2m}\sin 2\theta = 0$$

Method TWO:

$$\vec{r}_{A/O} = L\sin\theta\,\hat{i}\,, \qquad \vec{v}_{A/O} = L\cos\theta\,\dot{\theta}\,\hat{i}$$
$$\vec{a}_{A/O} = \left(-L\sin\theta\,\dot{\theta}^2 + L\cos\theta\,\ddot{\theta}\right)\hat{i}$$

$$\vec{r}_{B/O} = L\cos\theta\,\hat{j}\,, \qquad \vec{v}_{B/O} = -L\sin\theta\,\dot{\theta}\,\hat{j}$$
$$\vec{a}_{B/O} = \left(-L\cos\theta\,\dot{\theta}^2 - L\sin\theta\,\ddot{\theta}\right)\hat{j}$$

FBD for mass A:



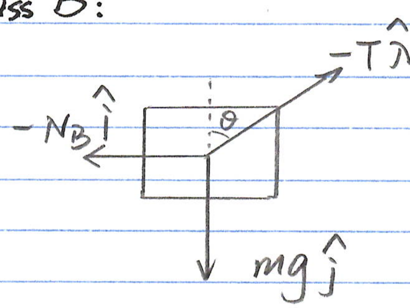
LMB:

$$-N_A\hat{j} + mg\hat{j} - kL\sin\theta\,\hat{i} + T\hat{\lambda} = m\vec{a}_{A/O}$$

$\hat{i}\cdot\{\ \}$:

$$-kL\sin\theta - T\sin\theta = m\left(-L\sin\theta\,\dot{\theta}^2 + L\cos\theta\,\ddot{\theta}\right) \qquad (*)$$

FBD for mass B:



LMB:

$$-N_B\hat{i} - T\hat{\lambda} + mg\hat{j} = m\vec{a}_{B/O}$$

$\hat{j}\cdot\{\ \}$:

$$-T\cos\theta + mg = m\left(-L\cos\theta\,\dot{\theta}^2 - L\sin\theta\,\ddot{\theta}\right) \qquad (**)$$

$(*) \times \cos\theta \quad - (**) \times \sin\theta :$

$$- kL\sin\theta\cos\theta - \cancel{T\sin\theta\cos\theta} + \cancel{T\cos\theta\sin\theta} - mg\sin\theta$$

$$= m(-L\sin\theta\cos\theta\,\dot{\theta}^2 + L\cos^2\theta\,\ddot{\theta})$$

$$- m(-L\cos\theta\sin\theta\,\dot{\theta}^2 - L\sin^2\theta\,\ddot{\theta})$$

$$\Rightarrow \quad - kL\sin\theta\cos\theta - mg\sin\theta$$

$$= mL\ddot{\theta}$$

i.e.,

$$\ddot{\theta} + \frac{g}{L}\sin\theta + \frac{k}{2m}\sin 2\theta = 0$$

same as Method ONE.